

DESIGN AND DEVELOPMENT OF A CONTAINERIZED COLLEGE EVENT MANAGEMENT SYSTEM WITH CI/CD

A centralized college event platform with automated containerized deployment using Docker, Kubernetes, and CI/CD.

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ABSTRACT

The College Event Management System is a web-based application developed to provide a centralized platform for managing college events and student registrations. Students can register and log in to the system, view upcoming events, check details such as event date, time, venue and description, and register for their preferred events. Administrators can manage the complete event lifecycle by adding, updating, and deleting events and viewing the students registered for each event.

The system will be developed using React.js for the frontend, Node.js and Express.js for the backend, and MySQL for storing user, event, and registration data. The application will follow a modular architecture in which the frontend communicates with the backend through APIs, while the backend performs the required operations on the database. This provides a structured and centralized approach to handling college event information and registrations.

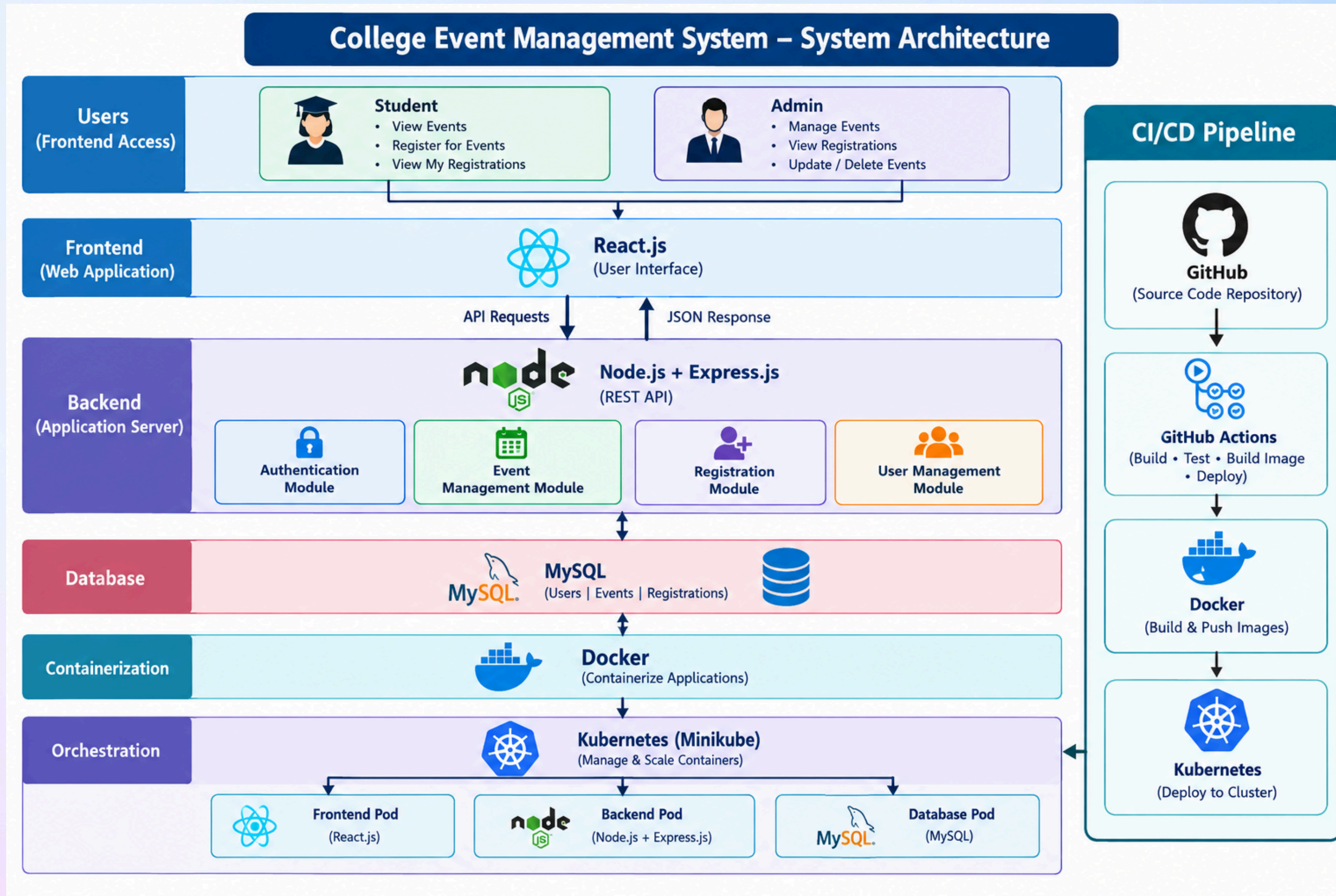
To incorporate DevOps concepts, the application will be containerized using Docker and managed using Kubernetes. Kubernetes will handle application deployment and container management. A CI/CD pipeline using GitHub Actions will automate the process of building, testing, creating Docker images, and deploying updated versions of the application. Thus, the project demonstrates not only a functional college event management system but also the practical use of Docker, Kubernetes, version control, and CI/CD automation for efficient application deployment and management.

LITERATURE SURVEY

Literature Survey

Sr. No.	Authors & Year	Title	Key Contribution	Relevance to Our Project
1	Kunal Arade, Omkar Shिवtare, Yash Patil & P. P. Patil, 2026	<i>Eventcraft: College Event Management System</i>	Developed a digital college event management system using the MERN stack for managing academic and event activities, including event management and student participation.	Supports the core event-management functionality, especially digital event handling and student participation.
2	Gaurav Thombare, Pavan Jadhav, Sachin Dhere, Prasad Jadhav & Priyanka Raikar, 2024	<i>Evolving Trends in College Event Management: A Comprehensive Survey and Implementation Analysis</i>	Studied the transition from traditional event-management methods toward modern digital platforms in educational institutions.	Supports replacing manual event management with a centralized digital platform.
3	Aditi Chaturvedi, Krishna Sharma, Akshat Dua & Aastha Gupta, 2024	<i>CU-EVENTS: A Comprehensive Event Management System for University</i>	Introduced a three-layer architecture with event organization, registration and participation tracking.	Supports our frontend-backend-database architecture and event registration functionality.
4	A. Alamoush & H. Eichelberger, 2024	<i>Open Source Container Orchestration for Industry 4.0 - Requirements and Systematic Feature Analysis</i>	Analyzed open-source container orchestration platforms and their features, including Kubernetes.	Supports our use of Kubernetes for container orchestration, management and scalability.
5	Ricardo Amaro, Rúben Pereira & Miguel Mira da Silva, 2025	<i>Mapping DevOps Capabilities to the Software Life Cycle: A Systematic Literature Review</i>	Studied DevOps capabilities such as CI/CD, version control, containers, microservices and cloud-native practices across the software lifecycle.	Provides the DevOps foundation for our Docker, Kubernetes and CI/CD architecture.
6	Aby Thankachan, Muhammed Bin Shafeeq T.P., Shanthanu S. Nair, Shijin Kumar, Sreethi S. & Shameem Ansar, 2025	<i>Design and Implementation of a Comprehensive College Event Management System</i>	Presented a unified college event-management system supporting event management, student participation, resource allocation and administrative activities.	Supports our event-management, student participation and administrative modules.

SYSTEM ARCHITECTURE



The background features a soft, abstract pattern of pastel brushstrokes in shades of pink, purple, and blue. Overlaid on this are several large, dark blue geometric shapes, primarily triangles and parallelograms, which are positioned in the corners and along the right edge, creating a modern, layered effect.

THANK YOU